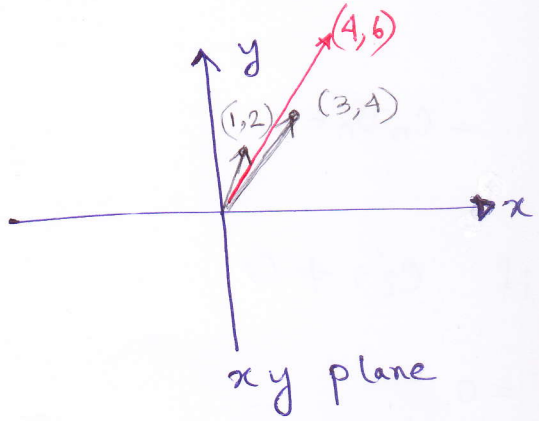
MAT 216
Week 5
Lec 9 – "highlights"

{ Watch the Video Lecture and Read the Lec NOTE given in buX }

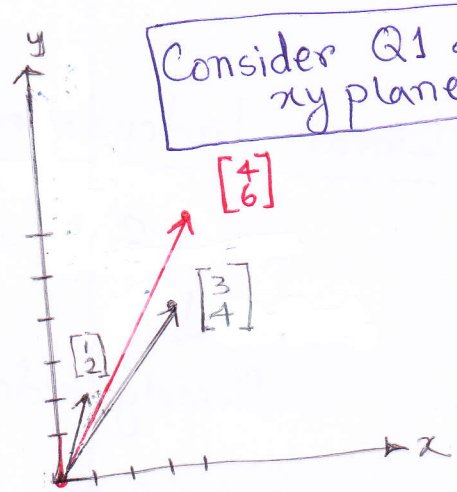
Vector Space

Vector spaces are collections of vectors that follow certain rules and properties.

(*) All addition of vectors in a given vector space and multiplication with any scalar quantity must produce vectors that are also present in the same vector space.



Consider Q1 & xy plane

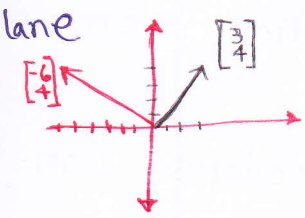


$$\begin{bmatrix}3\\4\end{bmatrix} + \begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}4\\6\end{bmatrix}$$

[both vectors are in Q1 & their addition is also in Q1

Q1 xy plane — Q1 xy plane — Q1 xy Plane

$$(-2)\begin{bmatrix}3\\4\end{bmatrix} \rightarrow \begin{bmatrix}-6\\4\end{bmatrix}$$



[the vector is in Q1 but its scalar multiplication is not in Q1]

∴ Q1 in not a Vector Space

xy-plane represent $\mathbb{R}^2$ (2-dimensional vector Space)

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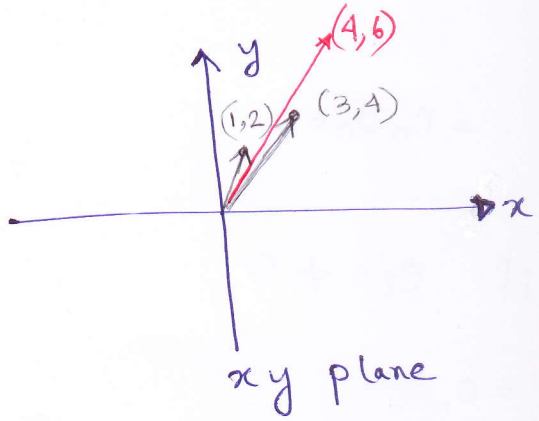

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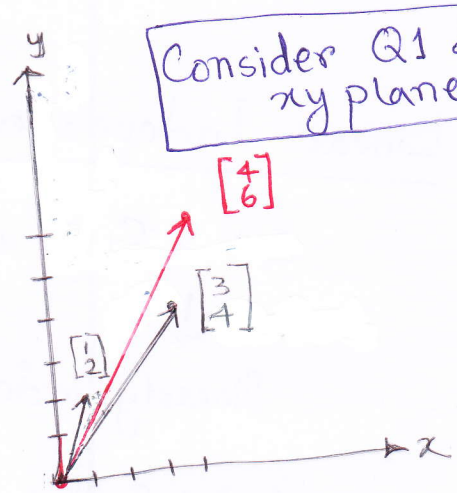

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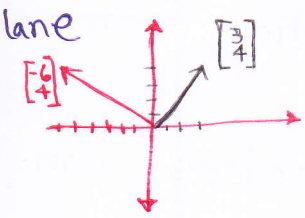

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xy-plane represent $\mathbb{R}^2$ (2-dimensional vector Space)

(**) $\vec{0} \in$ All vector space

↳ zero vector

Few examples:

R^3 gives the Vector Space containing all vectors in xyz plane: $\begin{bmatrix} 0 \\ \pi \\ e \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ 6 \end{bmatrix}$

R^6 gives the V.S. with 6 dimensional vectors with real components: $\begin{bmatrix} 9 \\ 1 \\ 0 \\ 0 \\ 5 \\ -1 \end{bmatrix}$

Linear Independence:

$$c_1 v_1 + c_1 v_2 + c_3 v_3 + \cdots + c_n v_n = 0$$

↓

linearly independent if c_i's $\neq 0$

$$c_1 v_1 + c_2 v_2 + \cdots + c_n v_n = 0$$

↓

linearly dependent if at least one of the "c" = 0. Since we can always make the combination zero with the presence of a zero vector in it.

Vectors can also be multiples of each other

$$v_1 = -c v_2$$
$$\Rightarrow v_1 + c v_2 = 0$$
$$\Rightarrow 0 \cdot v_1 + c \cdot 0 = 0$$

Such vectors can always be expressed as linear combinations of the other.

→ linearly dependent.

Spanning a Space

$$\text{span}(\underbrace{v_1, v_2 \ldots, v_n}_{\text{set of vectors}}) = \underbrace{c_1 v_1 + c_2 v_2 + \cdots + c_n v_n}_{\text{all possible linear Combination of the vector in that set}}$$

If the spanning set of vectors i.e. $\{v_1, v_2, \ldots, v_n\}$ produces all the vectors in any vector space V, then we say that the vectors span the space:

$$\text{Space, } V = \text{Span}\{v_1, v_2, \ldots, v_n\}$$

Show $\{(1,2,0), (0,5,7), (-1,1,3)\}$ span the V.S. $\mathbb{R}^3$

- Let Space $V = (b_1, b_2, b_3)$

$$(b_1, b_2, b_3) = k_1(1,2,0) + k_2(0,5,7) + k_3(-1,1,3)$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & b_1 \\ 2 & 5 & 1 & b_2 \\ 0 & 7 & 3 & b_3 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & -1 & b_1 \\ 0 & 5 & 3 & b_2-2b_1 \\ 0 & 7 & 3 & b_3 \end{array}\right] R_2 - 2R_1$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & b_1 \\ 0 & 5 & 3 & b_2-2b_1 \\ 0 & 0 & -6 & 5b_3-7b_2+14b_1 \end{array}\right] 5R_3 - 7R_2$$

$$= \left[\begin{array}{ccc|c} 6 & 0 & 0 & 6b_1-5b_3+7b_2-14b_1 \\ 0 & 10 & 0 & 2b_2-4b_1+5b_3-7b_2+14b_1 \\ 0 & 0 & -6 & 5b_3-7b_2+14b_1 \end{array}\right] \begin{array}{l} 6R_1 - R_3 \\ 2R_2 + R_3 \\ \end{array}$$

$$= \left[\begin{array}{ccc|c} 6 & 0 & 0 & -8b_1 + 7b_2 - 5b_3 \\ 0 & 10 & 0 & 10b_1 - 5b_2 + 5b_3 \\ 0 & 0 & -6 & 10b_1 - 7b_2 + 5b_3 \end{array}\right]$$

$$6k_1 = -8b_1 + 7b_2 - 5b_3$$

$$k_1 = \frac{1}{6}(-8b_1 + 7b_2 - 5b_3) = \frac{1}{6}\begin{bmatrix} -8b_1 \\ 7b_2 \\ -5b_3 \end{bmatrix}$$

$$k_2 = \frac{1}{10}\begin{bmatrix} 10b_1 \\ -5b_2 \\ 5b_3 \end{bmatrix} \qquad k_3 = -\frac{1}{6}\begin{bmatrix} 10b_1 \\ -7b_2 \\ 5b_3 \end{bmatrix}$$

Bases for the space $\mathbb{R}^3$

$$\begin{bmatrix} -4/3 \\ 7/6 \\ -5/6 \end{bmatrix}, \begin{bmatrix} 1 \\ -1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} -5/3 \\ 7/6 \\ -5/6 \end{bmatrix}$$

No need to evaluate in this way
Just showing the calculation to make the concept clear.

Example: Possible bases for $\mathbb{R}^3$

- $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
- $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 7 \end{bmatrix}$

We are choosing ~~the~~ basis differently, the number of vectors in each choice is fixed.

Summary of all properties of a V.S.

- Commutativity: $u+v=v+u$
- Associativity: $(u+v)+w=u+(v+w)$
- Additive Identity: $u+0=u$
- Additive Inverse: $u+(-u)=0$
- Multiplicative Inverse: $1u=u$
- Associativity of scalar multiplication: $(c_1 c_2)u=c_1(c_2 u)$
- Distributive property of scalar multiplication

1. $c(u+v)=cu+cv$
2. $(c_1+c_2)u=c_1u+c_2u$

If S is any set of vectors in a Vector Space:

- $u, v \in S$ and $\begin{cases} u+v \in S \\ cu \in S \\ \bullet\ c \text{ is a scalar no.} \end{cases}$ Then S is a V. Subspace

Consider the example we had in page 1 of Vector space.

V. Subspace has the same rules as vector space.

Let's take a span of a single vector $\begin{bmatrix} a \\ b \end{bmatrix}$

Span $\begin{bmatrix} a \\ b \end{bmatrix} = c\begin{bmatrix} a \\ b \end{bmatrix} \longrightarrow$

- $c_1\begin{bmatrix} a \\ b \end{bmatrix} + c_2\begin{bmatrix} a \\ b \end{bmatrix}$
- $(c_1+c_2)\begin{bmatrix} a \\ b \end{bmatrix}$

Thus it also satisfies the properties of subspace.

Span of two or more vectors will form a subspace

$\text{span}(a, b) = c_1 a + c_2 b$

- zero vector : $0 \cdot a + 0 \cdot b = 0$ [check for existence of zero vectors]
- take 2 vectors x & y and check addition property

$x = c_1 a + c_2 b$

$y = d_1 a + d_2 b$

Vector Addition: $x + y = (c_1 + d_1) a + (c_2 + d_2) b$

- <u>Scalar Multiplication:</u> S.M. will give another combination in the span.

<u>Subspace inside $\mathbb{R}^3$:</u>

- zero vector itself: $0 \cdot a + 0 \cdot b + 0 \cdot c = 0$

or $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

- Any line through the origin $c \begin{bmatrix} a \\ b \\ c \end{bmatrix}$
- Any plane through the origin $c_1 \begin{bmatrix} a \\ b \\ c \end{bmatrix} + c_2 \begin{bmatrix} d \\ e \\ f \end{bmatrix}$
- The vector space $= \mathbb{R}^3$

(The V.S. is $\mathbb{R}^3$ itself)

Matrix Space:

Consider a set of $m \times n$ matrices in this matrix

Considering the set of all 3×3 matrices

- Addition of 3×3 matrices is also 3×3 matrix.

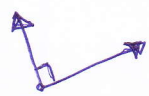
- If we multiply a vector in this set with a real number, we will get another vector in this set, thus a 3×3 matrix

$$p \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \text{another } 3 \times 3 \text{ matrix.}$$

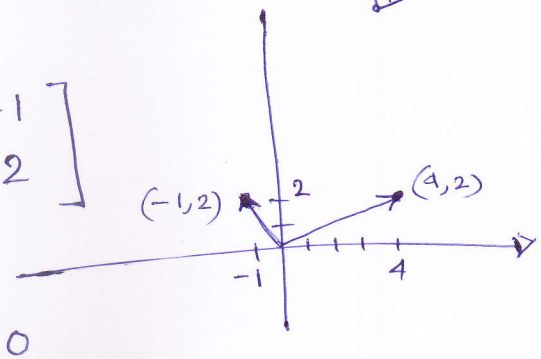
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \text{another } 3 \times 3 \text{ matrix}$$

Orthogonality of Subspace

If the dot product of any 2 vectors are "0", then they are orthogonal to each other.



$$u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad v = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$



$$u \cdot v = 0$$

$$(4, 2) \cdot (-1, 2) = 0$$

$\therefore$ u & v are orthogonal.

If this is true for any two vectors such as $u \in U$, $v \in V$ then the respective subspaces will be called orthogonal subspaces. So, U and V are orthogonal subspaces here.